HARD · ADVANCED MATH · ANSWER KEY

Advanced Math

07

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

D

D. $-\frac{5}{2}$

Choice D is correct. It's given that $f(x) = (x - 14)(x + 19)$, which can be rewritten as $f(x) = x^2 + 5x - 266$. Since the coefficient of the x^2-term is positive, the graph of $y = f(x)$ in the xy-plane opens upward and reaches its minimum value at its vertex. The x-coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x) = ax^2 + bx + c$, where a, b, and c are constants, the x-coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 + 5x - 266$, $a = 1$, $b = 5$, and $c = -266$. It follows that the x-coordinate of the vertex is $-\frac{5}{2(1)}$, or $-\frac{5}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x-value of the midpoint between the two x-intercepts of the parabola. Since it's given that $f(x) = (x - 14)(x + 19)$, it follows that the two x-intercepts of the graph of $y = f(x)$ in the xy-plane occur when $x = 14$ and $x = -19$, or at the points $(14, 0)$ and $(-19, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2), is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Therefore, the midpoint between $(14, 0)$ and $(-19, 0)$ is $\left(\frac{14 + (-19)}{2}, \frac{0 + 0}{2}\right)$, or $\left(-\frac{5}{2}, 0\right)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Choice A is incorrect. This is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane.

Choice B is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = f(x)$ in the xy-plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2 C

C. There is exactly 1 solution.

Choice C is correct. The second equation of the system can be rewritten as $y=5x-8$. Substituting $5x-8$ for y in the first equation gives $5x-8=x^2+3x-7$. This equation can be solved as shown below:

$x^2+3x-7-5x+8=0$

$x^2-2x+1=0$

$(x-1)^2=0$

$x=1$

Substituting 1 for x in the equation $y=5x-8$ gives $y=-3$. Therefore, $(1,-3)$ is the only solution to the system of equations.

Choice A is incorrect. In the xy-plane, a parabola and a line can intersect at no more than two points. Since the graph of the first equation is a parabola and the graph of the second equation is a line, the system cannot have more than 2 solutions. Choice B is incorrect. There is a single ordered pair (x,y) that satisfies both equations of the system. Choice D is incorrect because the ordered pair $(1,-3)$ satisfies both equations of the system.

Q3 6632

The correct answer is 6632. Applying the distributive property to the expression yields $(7532+100y^2)+(100y^2-1100)$. Then adding together $7532+100y^2$ and $100y^2-1100$ and collecting like terms results in $200y^2+6432$. This is written in the form ay^2+b, where $a=200$ and $b=6432$. Therefore $a+b=200+6432=6632$.

Q4

D

D. The estimated number of advertisements the company sent to its clients in 1997 was 9,000.

Choice D is correct. The y-intercept of a graph in the xy-plane is the point where $x = 0$. For the given function f, the y-intercept of the graph of $y = fx$ in the xy-plane can be found by substituting 0 for x in the equation $y = 9{,}0000.66^x$, which gives $y = 9{,}0000.66^0$. This is equivalent to $y = 9{,}0001$, or $y = 9{,}000$. Therefore, the y-intercept of the graph of $y = fx$ is $0, 9{,}000$. It’s given that the function f models the number of advertisements a company sent to its clients each year. Therefore, fx represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since 1997. Therefore, $x = 0$ represents 0 years since 1997, or 1997. Thus, the best interpretation of the y-intercept of the graph of $y = fx$ is that the estimated number of advertisements the company sent to its clients in 1997 was 9,000.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5

D

D. 40

Choice D is correct. Adding 10 to each side of the given equation yields $x^2 - 40x = 10$. To complete the square, adding $\frac{40}{2}^2$, or 20^2, to each side of this equation yields $x^2 - 40x + 20^2 = 10 + 20^2$, or $x - 20^2 = 410$. Taking the square root of each side of this equation yields $x - 20 = \pm\sqrt{410}$. Adding 20 to each side of this equation yields $x = 20 \pm \sqrt{410}$. Therefore, the solutions to the given equation are $x = 20 + \sqrt{410}$ and $x = 20 - \sqrt{410}$. The sum of these solutions is $20 + \sqrt{410} + 20 - \sqrt{410}$, or 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6

D

D. $\frac{9}{x + 14x - 5}$

Choice D is correct. The expression $\frac{4}{4x - 5} - \frac{1}{x + 1}$ can be rewritten as $\frac{4}{4x - 5} + \frac{-1}{x + 1}$. To add the two terms of this expression, the terms can be rewritten with a common denominator. Since $\frac{x + 1}{x + 1} = 1$, the expression $\frac{4}{4x - 5}$ can be rewritten as $\frac{x + 14}{x + 14x - 5}$. Since $\frac{4x - 5}{4x - 5} = 1$, the expression $\frac{-1}{x + 1}$ can be rewritten as $\frac{4x - 5\text{-}1}{4x - 5x + 1}$. Therefore, the expression $\frac{4}{4x - 5} + \frac{-1}{x + 1}$ can be rewritten as $\frac{x + 14}{x + 14x - 5} + \frac{4x - 5\text{-}1}{4x - 5x + 1}$, which is equivalent to $\frac{x + 14 + 4x - 5\text{-}1}{x + 14x - 5}$. Applying the distributive property to each term of the numerator yields $\frac{4x + 4 + -4x + 5}{x + 14x - 5}$, or $\frac{4x + -4x + 4 + 5}{x + 14x - 5}$. Adding like terms in the numerator yields $\frac{9}{x + 14x - 5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7

D

D. The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

Choice D is correct. The y-intercept of a graph in the ty-plane is the point where $t = 0$. For the given function f, the y-intercept of the graph of $y = ft$ in the ty-plane can be found by substituting 0 for t in the equation $y = 8{,}0000.65^t$, which gives $y = 8{,}0000.65^0$. This is equivalent to $y = 8{,}0001$, or $y = 8{,}000$. Therefore, the y-intercept of the graph of $y = ft$ is $0, 8{,}000$. It's given that the function f models the number of coupons a company sent to their customers at the end of each year. Therefore, ft represents the estimated number of coupons the company sent to their customers at the end of each year. It's also given that t represents the number of years since the end of 1998. Therefore, $t = 0$ represents 0 years since the end of 1998, or the end of 1998. Thus, the best interpretation of the y-intercept of the graph of $y = ft$ is that the estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8

20.25

Accept also: 81/4

The correct answer is $\frac{81}{4}$. The given linear equation is $2y = c$. Dividing both sides of this equation by 2 yields $y = \frac{c}{2}$. Substituting $\frac{c}{2}$ for y in the equation of the parabola yields $\frac{c}{2} = -2x^2 + 9x$. Adding $2x^2$ and $-9x$ to both sides of this equation yields $2x^2 - 9x + \frac{c}{2} = 0$. Since it's given that the line and the parabola intersect at exactly one point, the equation $2x^2 - 9x + \frac{c}{2} = 0$ must have exactly one solution. An equation of the form $Ax^2 + Bx + C = 0$, where A, B, and C are constants, has exactly one solution when the discriminant, $B^2 - 4AC$, is equal to 0. In the equation $2x^2 - 9x + \frac{c}{2} = 0$, where $A = 2$, $B = -9$, and $C = \frac{c}{2}$, the discriminant is $-9^2 - 42\frac{c}{2}$. Setting the discriminant equal to 0 yields $-9^2 - 42\frac{c}{2} = 0$, or $81 - 4c = 0$. Adding $4c$ to both sides of this equation yields $81 = 4c$. Dividing both sides of this equation by 4 yields $c = \frac{81}{4}$. Note that 81/4 and 20.25 are examples of ways to enter a correct answer.

Q9

D

D. 29

Choice D is correct. If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, then the expression can be written as $k - xax + b$, where a and b are constants. This expression can be rewritten as $akx + bk - ax^2 - bx$, or $-ax^2 + ak - bx + bk$. Since this expression is equivalent to $-x^2 + \frac{1}{29}nk^2$, it follows that $-a = -1$, $ak - b = 0$, and $bk = \frac{1}{29}nk^2$. Dividing each side of the equation $-a = -1$ by -1 yields $a = 1$. Substituting 1 for a in the equation $ak - b = 0$ yields $k - b = 0$. Adding b to each side of this equation yields $k = b$. Substituting k for b in the equation $bk = \frac{1}{29}nk^2$ yields $k^2 = \frac{1}{29}nk^2$. Since k is positive, dividing each side of this equation by k^2 yields $1 = \frac{1}{29}n$. Multiplying each side of this equation by 29 yields $29 = n$.

Alternate approach: The expression $x^2 - y^2$ can be written as $x - yx + y$, which is a difference of two squares. It follows that $\frac{1}{29}nk^2 - x^2$ is equivalent to $\sqrt{\frac{1}{29}n}\,k - x\sqrt{\frac{1}{29}n}\,k + x$. It's given that $k - x$ is a factor of $-x^2 + \frac{1}{29}nk^2$, so the factor $\sqrt{\frac{1}{29}n}\,k - x$ is equal to $k - x$. Adding x to both sides of the equation $\sqrt{\frac{1}{29}n}\,k - x = k - x$ yields $\sqrt{\frac{1}{29}n}\,k = k$. Since k is positive, dividing both sides of this equation by k yields $\sqrt{\frac{1}{29}n} = 1$. Squaring both sides of this equation yields $\frac{1}{29}n = 1$. Multiplying both sides of this equation by 29 yields $n = 29$.

Choice A is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}-29k^2$, or $-x^2 - k^2$. This expression doesn't have $k - x$ as a factor.

Choice B is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}-\frac{1}{29}k^2$, or $-x^2 + -\frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Choice C is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}\frac{1}{29}k^2$, or $-x^2 + \frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Q10 D

D. Neither I nor II

Choice D is correct. It's given that the graph of $y = fx$ in the xy-plane is a parabola with vertex h, k. If $f\text{-}9 = f3$, then for the graph of $y = fx$, the point with an x-coordinate of -9 and the point with an x-coordinate of 3 have the same y-coordinate. In the xy-plane, a parabola is a symmetric graph such that when two points have the same y-coordinate, these points are equidistant from the vertex, and the x-coordinate of the vertex is halfway between the x-coordinates of these two points. Therefore, for the graph of $y = fx$, the points with x-coordinates -9 and 3 are equidistant from the vertex, h, k, and h is halfway between -9 and 3. The value that is halfway between -9 and 3 is $\frac{-9+3}{2}$, or -3. Therefore, $h = -3$. The equation defining f can also be written in vertex form, $fx = ax - h^2 + k$. Substituting -3 for h in this equation yields $fx = ax - -3^2 + k$, or $fx = ax + 3^2 + k$. This equation is equivalent to $fx = ax^2 + 6x + 9 + k$, or $fx = ax^2 + 6ax + 9a + k$. Since $fx = ax^2 + 4x + c$, it follows that $6a = 4$ and $9a + k = c$. Dividing both sides of the equation $6a = 4$ by 6 yields $a = \frac{4}{6}$, or $a = \frac{2}{3}$. Since $\frac{2}{3} < 1$, it's not true that $a \geq 1$. Therefore, statement II isn't true. Substituting $\frac{2}{3}$ for a in the equation $9a + k = c$ yields $9\frac{2}{3} + k = c$, or $6 + k = c$. Subtracting 6 from both sides of this equation yields $k = c - 6$. If $k < 0$, then $c - 6 < 0$, or $c < 6$. Since c could be any value less than 6, it's not necessarily true that $c < 0$. Therefore, statement I isn't necessarily true. Thus, neither I nor II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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